

Assignment-3

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Branch: B-Tech CSE with AI & ML

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Subject: Software Engineering

① Explain object-oriented vs Function-oriented Design.

→ (OOD)

- object-oriented design focuses on objects and classes. It combines data and functions into a single unit called an object.

Point:

- Emphasis on data & behaviour together
- Uses bottom-up approach
- Supports encapsulation, inheritance, polymorphism
- Better data security
- Examples: Java, C++, Python

(FOD)

- Function-oriented design focuses on functions (procedures). The system is divided into smaller functions, and each function performs a specific task. Data is passed between functions.

Points:

- Emphasis on Process / Logic
- uses top-down approach
- Functions are independent but share data
- Data is less secure
- Examples: C, Pascal

② Discuss Architectural Design concepts.

→ Architectural design in software engineering refers to the high-level structure of a system, showing how different components interact and are organized.

It acts as a blueprint that defines:

- System structure
- components (modules)
- Relationships and communication.

Key Architectural Design concepts

1) Abstraction

↳ focuses on essential features, ignoring unnecessary details

ii) Modularity

↳ Divides the system into smaller independent modules

III) Encapsulation

↳ Hides internal details of a module

IV) Separation of concerns

↳ Each part of the system handles a specific responsibility

V) Data structure

↳ organizes how data is stored and accessed

VI) Software Architectural styles

- Client-server
- Layered Architecture
- Microservices

③ Explain Data and Behaviour Modeling.

→ Data modeling

- It focuses on how data is organized, and related in a system. It defines the structure of data and relationship between different data elements.

Points:

- Identifies entities (e.g., customer, Account)
- Defines attributes (e.g., name, balance)
- Shows relationships between entities
- Uses ER diagram.

Behaviour modeling

- It describes how the system behaves over time and how it responds to input or events.

Points:

- Focuses on dynamic behaviour of the system
- Shows data state changes and interactions
- Uses diagrams like:
 - ↳ State diagram
 - ↳ Sequence diagram
 - ↳ Activity diagram
- Helps in understanding system functionality and flow.

④ Analyze the importance of modularity in design.

→ Modularity is a design concept in which a software system is divided into smaller, independent units called modules, each performing a specific function.

It helps in managing complexity by breaking a large system into manageable and organized parts.

Importance / advantages of modularity

- i) Reduced complexity
- ii) Easy maintenance
- iii) Improved Reusability

- iv) Better testing and debugging
- v) Parallel Development
- vi) High cohesion and low coupling
- vii) Scalability
- viii) Improved Reliability

⑤ Explain Design Document Structure

→ A Design Document (also called Software Design Document) describes the overall architectural and detailed design of a Software System.

It acts as a blueprint for developers, explaining how the system will be built based on the requirements.

Typical Structure of a Design Document

i) Introduction

- purpose of the document
- Scope of the system

ii) System overview

- General description of the system
- major functionalities

iii) Architectural Design

- overall system architecture
- interaction between components

iv) Data Design

- Data structure and database design
- ER diagrams, tables, relationships

v) Interface Design

vi) Component level diagram design

- Detailed description of each module
- Algorithms, logic,

vii) Security design

- Authentication and authorization mechanism
- Risk and security measures

viii) Performance and constraints

- Hardware / software requirements
- system limitations

ix) Testing considerations

- testing strategies
- test case and validation methods.

⑥ Discuss coding styles and best practices.

→ Coding styles refers to a set of guidelines and conventions used while writing code to make it readable, consistent, and maintainable.

Best practices are proven methods that help developers write efficient, error-free, and high-quality code.

■ Coding styles

- i) meaningful naming
- ii) Proper indentation
- iii) code formatting
- iv) Comments and documentation

■ Best Practices

- i) write simple and clear code
- ii) Avoid code duplication
- iii) Error Handling
- iv) code Reusability
- v) testing and debugging
- vi) Follow standards
- viii) version control

⑦ Design a simple system architecture

→ A Library management System is designed to manage books, users, and transactions such as issuing and returning books.

The architecture defines how different components interact to perform these operations efficiently.

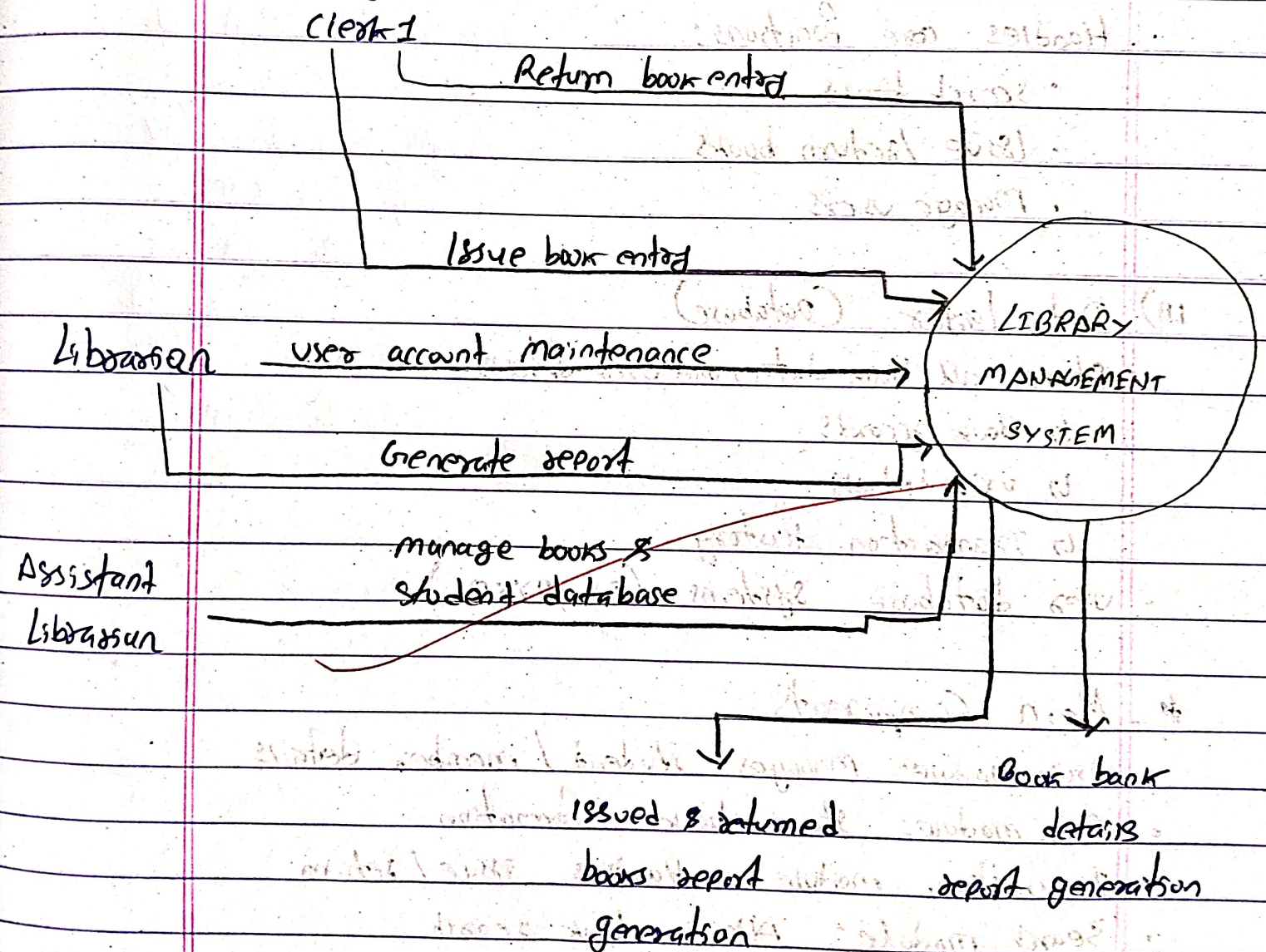


Fig: Data Flow Diagram

Layered approach

i) Presentation layer (UI)

- Interface for users (student, librarian)
- Display book details, issue/return options
- Accept user inputs

ii) Application layer (Logic)

- Processes user requests
- Handles core functions:
 - Search books
 - Issue / return books
 - Manage users

iii) Data layer (Database)

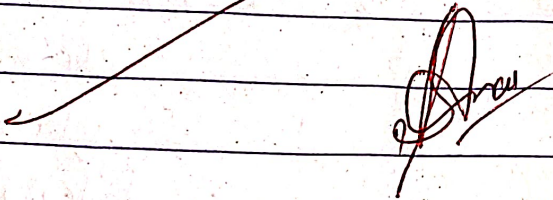
- Stores all book data, and user details
 - ↳ Book records
 - ↳ User details
 - ↳ Transaction History
- Uses database systems (eg. MySQL)

* Main Components

- User module: Manages student / member details
- Book module: stores book information
- Transaction module: Handles issue / return
- Search module: Allows book search

* Working Flow

- i) user searches or requests a book via UI
- ii) Application layer processes request
- iii) Data is fetched / updated in database
- iv) Result is shown to the user

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